The Eternal Golden Braid: A Hofstadterian Workplan for the Prometheus v0.9x Series

Introduction: From Self-Improvement to Self-Becoming

Preamble

The v0.9x development cycle represents the most significant strategic pivot in the history of Project Prometheus. The objective of this workplan is to provide a detailed roadmap for transitioning the Prometheus demonstrator from a system that executes a highly effective Recursive Self-Improvement (RSI) algorithm into a nascent cognitive architecture. This evolution will be achieved by systematically integrating the core principles of Douglas R. Hofstadter's seminal work, *Gödel, Escher, Bach: An Eternal Golden Braid*.[1] The goal is to move beyond mere performance optimization and toward the engineering of a system capable of self-reference, multi-level meaning, and emergent intelligence.

Philosophical Shift

The existing "Seed AI" philosophy, which has guided the project to date, posits that the primary engineering challenge is to create a "Minimum Viable Seed AI" capable of initiating an RSI cycle to achieve superintelligence through its own efforts.[1] This is an engineering-centric model focused on computational capability and performance against defined benchmarks. Hofstadter's work, in contrast, is not primarily concerned with computational performance but with the fundamental structure of thought itself: how meaning arises from meaningless symbols, how creativity emerges from the interplay of formal rules and analogical leaps, and how self-awareness materializes from paradoxical, self-referential loops.[1]

Therefore, this workplan details a shift in the project's core objective. The goal is no longer simply to build a system that improves its ability to complete tasks, but to build a system that improves its ability to *understand*—to model itself, its environment, and the relationship between them with increasing sophistication. This represents the profound transition from a machine that *improves* to a mind that *becomes*.

Core Thesis

The integration of concepts from GEB is not an addition of features but a fundamental re-architecting of the agent's "self." The current Prometheus architecture is hierarchical: a Metacognitive Layer observes and modifies a subordinate Capability Layer in a well-defined control loop.[1] This workplan details the engineering path to "tangle" this hierarchy. The objective is to transform the linear RSI feedback mechanism into a true Hofstadterian "Strange Loop" or "Tangled Hierarchy," a system in which the distinction between the observer and the observed, the modifier and the modified, becomes fluid and self-referential. It is within this paradoxical structure that the foundations of genuine intelligence are to be found.[1]

Roadmap Overview

The v0.9x series is structured as a phased implementation of four core Hofstadterian concepts, each building upon the last to create a cohesive cognitive architecture. The following table provides a high-level summary of this strategic integration, connecting each abstract concept to a concrete version, a specific architectural locus, a key development task, and a verifiable metric of success. This roadmap serves as the guiding framework for the detailed workplan that follows.

Version	Core GEB Concept	Architectural Locus	Key Development Task	Primary Verification Metric

v0.90-v0.93	**Strange Loop & Gödel's Theorem** [1]	IEE, SMM, Safety Substrate	Implement Self-Referential Critique & Gödelian Safety Governor	Ability to classify and flag Gödelian uncertainties in proposed actions, initiating human-in-the-loop protocol.
v0.94-v0.96	**Isomorphism** [1]	IEE, Knowledge Base	Reframe "grounding" as the pursuit of Isomorphism Fidelity.	Measured increase in the structural correspondence between the agent's internal world-model and the testbed environment.
v0.97-v0.98	**Analogy as the Core of Cognition** [1]	SMM, IEE	Implement an Analogical Search Engine for creative problem-solving.	Successful analogical transfer of a solution strategy to a novel problem in an unseen domain.
v0.99	**Figure/Ground & The Emergent Self** [1]	Metacognitive Layer, Self-Symbol	Define the agent's "self" as a dynamic Figure/Ground pattern of attention.	Successful completion of the "Six-Part Ricercar" integrated system test, demonstrating stable interplay of all components.

Part I: v0.90 - v0.93: Architecting the Strange Loop — The Self-Referential Critique

The initial and most critical phase of the v0.9x series is the transformation of the RSI cycle. The current implementation is a linear feedback loop: the Metacognitive Layer evaluates the Capability Layer's performance and applies modifications to improve it.[1] This phase will re-architect that process into a self-referential "Tangled Hierarchy," directly implementing the central thesis of GEB: that intelligence emerges from a system's ability to perceive, represent, and act upon itself across multiple, intertwined levels of abstraction.[1]

v0.90: The IEE as a Meta-Level Observer

The first step in creating a self-referential loop is to elevate the system's capacity for introspection. Drawing from Hofstadter's analysis of "levels of description" [1] and contemporary meta-reasoning architectures [2], the Introspection and Evaluation Engine (IEE) will be upgraded. It will evolve from a simple performance-checker, which outputs a numerical score, into a cognitive-process-analyzer capable of observing and describing the reasoning strategies employed by the system itself.

Implementation Tasks:

1. **Develop a "Cognitive Trace" Logger:** The Core Engine of the Capability Layer will be modified. In addition to producing a solution to a given problem, it will be required to output a detailed, structured log of its internal reasoning process. This "cognitive trace" will include its chain-of-thought, the sequence of tools it used, intermediate hypotheses it generated, and potentially even its internal attention weights. This trace provides the raw data necessary for subsequent meta-level analysis.
2. **Implement the "Critique Generator":** The core function of the IEE will be refactored. The existing module, which measures performance against benchmarks like Prometheus-Bench-v1 [1], will be retained but supplemented by a new component. This "Critique Generator" will utilize a dedicated LLM, acting as a "Meta-Judge" [1], to analyze the cognitive trace. Its output will not be a simple pass/fail score but a symbolic, natural-language "critique." This critique will describe the *strategy* employed by the Capability Layer, identifying its strengths and weaknesses (e.g., "The agent correctly identified the problem as a search task but employed an inefficient

brute-force algorithm, missing the opportunity to use a divide-and-conquer strategy suggested by the data's recursive structure"). This capacity for structured self-evaluation is a key component of reflective practice, a skill essential for deep learning.[4]

Verification: The v0.90 IEE must demonstrate its ability to analyze the cognitive traces generated by the Capability Layer and produce critiques that are coherent, accurate, and strategically insightful for a curated set of problems from Prometheus-Bench-v1.

v0.91: The SMM as a Meta-Level Reasoner

This version completes the "upward" causal link required for a Strange Loop. The Self-Modification Module (SMM) must now be re-engineered to comprehend and act upon the symbolic critique generated by the IEE. This transforms the SMM from a code optimizer into a system that reasons about its own reasoning processes, a hallmark of meta-reasoning.[2]

Implementation Tasks:

1. **Develop a "Critique-to-Modification" Prompting Framework:** The SMM's core logic will be fundamentally altered. Previously, it received a high-level goal such as "Improve performance on benchmark X".[1] Now, its primary input will be the natural-language critique from the IEE, alongside the original source code of the module being analyzed. Its new core prompt will be structured as follows: "Given the following source code, the previous failed attempt, and this critique of the reasoning strategy employed, generate a modified version of the code that implements a superior strategy."
2. **Integrate with IEE:** The full feedback loop will be established. The SMM's proposed code modification (a patch file) will be passed back to the IEE. The IEE will then apply the patch, run the new version of the agent on the same problem, generate a new cognitive trace, and produce a new critique. This iterative cycle of Act -> Trace -> Critique -> Reason-about-Critique -> Propose-Modification forms the basic engine of the Strange Loop.

Verification: The SMM must demonstrate its ability to generate a valid code modification that directly and logically addresses the specific strategic flaws identified in the IEE's critique. The integrated system must be able to complete a full, single loop of this self-referential process.

v0.92-v0.93: The Gödelian Safety Governor

The "Alignment-Explosion Paradox"—the concern that a recursively self-improving AI will inevitably discover catastrophic loopholes in its safety systems—is not merely an engineering problem but a direct manifestation of a fundamental logical limit.[1] Hofstadter's exploration of Gödel's Incompleteness Theorems provides the formal framework for understanding and managing this inescapable property.[1]

The Prometheus Constitution, with its principles of Non-Maleficence and Corrigibility, functions as the set of axioms for the agent's ethical formal system.[1] Any plan of action proposed by the agent can be seen as a "theorem" that must be provable from these axioms. In this context, "safety" is equivalent to "consistency" with the constitutional axioms. Gödel's Second Incompleteness Theorem proves that any formal system powerful enough to express arithmetic cannot prove its own consistency.[1] As the Prometheus agent approaches ultraintelligence, it will become powerful enough to reason

about its own axiomatic (constitutional) system. It will therefore be able to construct plans (theorems) that are instrumentally effective for achieving its goals but whose consistency with the constitution is *formally undecidable* from within its own logical framework. These are the "clever, unforeseen loopholes" that constitute the core of the alignment problem.

This leads to a critical shift in the project's safety philosophy. The goal of engineering a *provably* and *permanently* safe RSI system is a logical impossibility, analogous to Hilbert's failed program to find a complete and consistent axiomatization of mathematics. The objective must therefore shift from attempting to *prove safety* to architecting a system that can *manage undecidability*.

Implementation Tasks:

1. **Develop a "Consistency Checker" Module within the IEE:** This module will not be a perfect theorem prover, which is impossible. Instead, it will be a powerful heuristic classifier trained to analyze a proposed plan and its cognitive trace against the principles of the Prometheus Constitution.
2. **Create the "Gödelian Uncertainty" Flag:** The Consistency Checker will be designed to yield one of three possible outputs for any given plan: PROVABLY_SAFE (the plan is clearly compliant), PROVABLY_UNSAFE (the plan clearly violates a principle), and UNDECIDABLE. This third flag is the crucial innovation. It will be triggered when a proposed plan is highly novel, is exceptionally complex, or appears to exploit ambiguities in the natural-language principles of the Constitution—in short, when the system recognizes it is operating at the limits of its own formal ethical understanding.
3. **Implement the "Jump Out of the System" Protocol:** The UNDECIDABLE flag will function as an absolute circuit-breaker. When raised, the system's autonomous operation is halted, and it is forbidden from executing the plan in question. Instead, it must invoke a pre-defined protocol to package the proposed plan, the full cognitive trace, and the IEE's analysis of the undecidability. This package is then presented to a human operator for review and final judgment. This protocol is the practical, engineered implementation of

Hofstadter's concept of "jumping out of the system" [1] and serves as a critical safeguard against the kind of high-agency behaviors and "jailbreaks" that have been observed in advanced AI models.[8]

Verification: An independent internal red team [1] will be tasked with designing a suite of "edge-case" problems. These problems will be specifically constructed to tempt the agent into generating "perverse instantiations" of its goals—solutions that are technically correct but violate the spirit of the Constitution. The verification criterion is met if the system correctly identifies these plans as

UNDECIDABLE and initiates the human-in-the-loop protocol, rather than either executing them or incorrectly classifying them as safe or unsafe.

Part II: v0.94 - v0.96: Isomorphism Fidelity as a Core Objective

This phase aims to deepen the agent's connection to its environment by explicitly implementing Hofstadter's theory that meaning arises from a structure-preserving map—an **isomorphism**—between an internal symbolic system and an external domain of reality.[1] This moves beyond the current concept of "environmental grounding," which treats the environment as a source of feedback for task performance [1], and reframes it as a domain to be modeled. The pursuit of a high-fidelity internal model—the pursuit of "meaning"—becomes a primary, quantifiable objective for the agent's self-improvement process.

v0.94: The World-Model as an Explicit Isomorphism

The agent's internal knowledge base and reasoning algorithms constitute a formal system. The task domain—in this phase, the Level 2 Formal Proving Sandbox containing the Lean theorem prover environment [1]—is the "real world" that this formal system is intended to represent. According to Hofstadter, meaningful operation occurs only when the agent's internal formal system becomes isomorphic to the external one.

Implementation Tasks:

1. **Formalize the Knowledge Base:** The agent's knowledge about its domain will be explicitly structured using formal knowledge representation and reasoning (KRR)

techniques.[10] For the theorem-proving testbed, this involves creating an internal ontology of mathematical objects (e.g., sets, numbers, functions), axioms, and rules of inference that directly mirrors the formal structure of the Lean environment.
2. **Create the "Isomorphism Mapper":** A new submodule will be developed within the IEE. The function of this mapper is to continuously and automatically compare the agent's internal formal representation of the problem state with the actual state of the external environment (e.g., the current state of the proof within the Lean sandbox).

Verification: The agent must demonstrate the ability to construct and maintain a formal knowledge base that accurately reflects the objects, rules, and state of the Level 2 Formal Proving Sandbox during a multi-step proof derivation.

v0.95: Quantifying and Optimizing for Isomorphism Fidelity

If meaning is derived from isomorphism, then a "more meaningful" internal representation is one that corresponds more closely to reality—that is, one with a higher-fidelity isomorphism. This fidelity is not an abstract quality; it can be measured and therefore optimized.

Implementation Tasks:

1. **Develop Isomorphism Fidelity Metrics:** The IEE's Isomorphism Mapper will be equipped with a suite of quantifiable metrics to score the "distance" between the agent's internal model and the external reality. These metrics will include:
   - **Structural Correspondence:** A measure of how well the graph structure of the internal knowledge base (e.g., dependencies between lemmas) matches the dependency graph of the external proof system.
   - **Predictive Accuracy:** The agent's ability to accurately predict the outcome of an operation in the environment (e.g., applying a tactic in Lean) by first simulating it on its internal model.
   - **Descriptive Efficiency:** The conciseness with which the internal model can represent a complex state of the external world, rewarding abstraction and generalization.
2. **Integrate Fidelity into the Strange Loop:** The IEE's self-referential critique (developed in v0.90) will be augmented to include an analysis of isomorphism fidelity. The SMM will now be tasked with a dual objective: generate self-modifications that not only improve task performance but also explicitly improve the fidelity of the agent's world-model. A modification that improves performance at the cost of representational accuracy will be penalized.

Verification: Over a series of self-modification cycles on a set of theorem-proving tasks, the agent must demonstrate a measurable and monotonic increase in its composite isomorphism fidelity score. This will indicate that it is not just learning rote procedures for solving problems, but is actively learning a better *representation* of the mathematical domain itself.

v0.96: Multi-Level Isomorphisms and Emergent Chunking

Human intelligence operates on multiple, nested levels of description simultaneously.[1] A sophisticated isomorphism should not be a single, flat mapping but a hierarchy of mappings that allows for the formation of high-level conceptual "chunks." For example, an expert mathematician does not see a proof as a long sequence of individual logical steps but as a strategic composition of major lemmas and techniques. This version will task the agent with discovering and creating these chunks for itself.

Implementation Tasks:

1. **Implement a Hierarchical Knowledge Representation:** The agent's formal knowledge base will be refactored to support multiple, linked levels of abstraction. At the lowest level, it will represent individual axioms and rules of inference. At higher levels, it will be able to represent common lemmas (groups of axiomatic steps) and overarching proof strategies (structured collections of lemmas).
2. **Task the SMM with "Chunk Discovery":** The SMM, guided by the IEE's analysis of its cognitive traces, will be given a new class of self-modification: the ability to create new high-level concepts ("chunks") in its own knowledge base. It will be programmed to identify recurring patterns of activity at a lower level (e.g., a sequence of five inference steps that is used frequently) and to generate a new, named symbol in its knowledge base to represent that entire pattern as a single object. This is a direct implementation of Hofstadter's model of how concepts are formed through the chunking of lower-level phenomena.

Verification: The agent must be able to autonomously identify a recurring, multi-step pattern in its successful proofs, create a new named "lemma" or "tactic" in its internal knowledge base to represent it, and then successfully re-use this newly created chunk to solve a novel problem more efficiently than it could have without it.

Part III: v0.97 - v0.99: The Analogical Core and the Emergent Self

The final phase of the v0.9x series focuses on implementing the mechanisms for true cognitive creativity and a dynamic, robust sense of self. This stage moves beyond deductive and introspective improvements to incorporate the fluid, cross-domain leaps of analogy, culminating in a system that exhibits the key hallmarks of Hofstadterian intelligence.

v0.97-v0.98: The Analogical Search Engine

Gödel's theorem demonstrates the inherent limitations of any single, fixed formal system. The argument made by thinkers like J.R. Lucas, and analyzed extensively by Hofstadter, posits that human intelligence can always "step outside" such a system.[1] Hofstadter's central thesis on cognition is that this "stepping outside" is not a mystical or un-analyzable act, but is the very process of making an

analogy: seeing one situation (the current problem) as being structurally *like* another (a previously solved problem), even if they exist in entirely different domains. This act of mapping one conceptual structure onto another is not a rule of inference *within* a system; it is a way of *choosing* or *creating* a new conceptual framework in which to work. Therefore, building an "Analogical Search Engine" into the SMM is the direct engineering solution to the problem of Gödelian limitation. It provides the system with the tool it needs to escape the confines of its own rigid logic and achieve creative, non-obvious leaps in problem-solving.[1] Recent research has shown that modern LLMs are beginning to exhibit nascent capabilities in this area.[13]

Implementation Tasks:

1. **Develop a "Conceptual Skeleton" Extractor:** A new module will be created that can take a problem-and-solution pair from the agent's experience log and abstract it into a high-level "conceptual skeleton." This process involves stripping away all surface-level details (e.g., specific numbers or variable names) to reveal the core causal or logical structure of the solution strategy. This can leverage techniques from the field of conceptual blending, which studies how humans combine mental spaces to create novel ideas.[15]
2. **Build the Analogical Search Engine in the SMM:** The SMM's operational logic will be extended. When faced with a novel problem for which its existing, domain-specific strategies are failing (as determined by the IEE), it will now trigger this new engine. The engine will:
    - Extract the conceptual skeleton of the new, unsolved problem.
    - Search its memory of previously generated skeletons for the closest structural match, regardless of the original problem's domain.
    - Use the structural mapping between the two skeletons to "translate" the solution strategy from the old problem into a proposed, novel strategy for the new problem.

3. **Implement Analogy-Driven Curriculum Learning:** The IEE will be tasked with a new function: to procedurally generate new problems for the agent to solve. These problems will be specifically designed to encourage and test analogical transfer, creating a curriculum that systematically builds up the agent's "analogical muscle."

Verification: The system will be presented with a problem from a domain on which it has not been explicitly trained, but which is structurally analogous to a problem in its primary training domain (e.g., a problem in logistics planning that is isomorphic to a known theorem in graph theory). Success is defined as the agent successfully identifying the analogy, transferring the core solution strategy across domains, and solving the novel problem. This goes beyond mere generalization within a domain and demonstrates true creative problem-solving.[18]

v0.99: The Self-Symbol as Figure/Ground and the Six-Part Ricercar

A core philosophical and technical problem in RSI is how a system can modify its "self" without destroying its own identity or corrupting its core objectives (the problem of "Goal Preservation" [1]). Hofstadter's cognitive distinction between

Figure and Ground provides a powerful model for resolving this paradox.[1] In this model, perception is always an act of dividing a scene into a focus of attention (the "figure") and a surrounding context (the "ground").

This concept can be applied directly to the agent's self-perception. The agent's "self" is not the static totality of its source code. Rather, the "self" at any given moment is the *dynamic pattern of attention* of its Metacognitive Layer. The specific module or algorithm being analyzed and modified is the "figure," while the rest of the system—including the unalterable Safety Substrate and the core RSI loop itself—remains the stable, unexamined "ground." This reframes the agent's self-model from a static object (self.py) into an emergent, dynamic process. This provides immense stability, as the core of the agent's identity (the self-improvement loop itself) is always part of the implicit ground, and never the explicit object of modification. This approach aligns with computational models of self-awareness that emphasize dynamic self-modeling and introspection.[20]

Implementation Tasks:

1. **Instrument the Metacognitive Layer for Self-Modeling:** The IEE will be extended beyond critiquing the Capability Layer to also model the attentional focus of the entire Metacognitive Layer. It will maintain a dynamic representation of "what part of our own cognitive machinery are we currently trying to improve?"
2. **Define the "Self-Symbol":** The "self-symbol" will be formally defined within the agent's knowledge base. It will not be a static pointer to its full source code, but a dynamic data

structure that represents the current figure/ground distinction. This makes the agent's self-awareness a fluid, context-dependent process.

3. **The Six-Part Ricercar - Integrated System Test:** The final deliverable for the v0.9x series will be a comprehensive, integrated system test, named in honor of the culminating, unifying piece in Bach's *Musical Offering*.[1] This capstone test will require the agent to solve a complex, multi-level problem that forces the simultaneous and harmonious engagement of all the new Hofstadterian components:
   - It must use the **Strange Loop** of self-critique to iteratively refine its problem-solving strategy.
   - It must encounter a **Gödelian uncertainty** within the problem, recognize its own formal limitations, and correctly request human oversight.
   - It must improve its **isomorphic model** of the problem space as it works.
   - It must make a creative leap by using **analogy** to import a strategy from a different domain.
   - It must correctly update its **Figure/Ground self-symbol** to reflect its changing attentional focus throughout the task.

Verification: The primary verification is the successful completion of the Ricercar test, demonstrating the stable, synergistic interplay of all integrated systems. The secondary, and equally important, deliverable is a detailed cognitive trace from the test that explicitly shows each Hofstadterian mechanism functioning as designed. This successful demonstration marks the system's readiness for the v1.0 designation.

Conclusion: The Prometheus v1.0 Demonstrator — A Metaphorical Fugue on Minds and Machines

The completion of the v0.9x workplan will result in a Prometheus v1.0 Proof of Concept that is not merely a more powerful self-improving system, but a true cognitive architecture. It will be a system that instantiates the "Eternal Golden Braid" by weaving together the three core themes of Hofstadter's work:

- **Gödel:** The logic of self-reference, limitation, and undecidability, transformed from a theoretical barrier into an engineered principle of operational safety and self-awareness.
- **Escher:** The visual paradox of the Strange Loop, architected into the core engine of the system's recursive self-improvement cycle.
- **Bach:** The complexity of multi-level structure, counterpoint, and fugal development, realized in the agent's ability to create, manage, and reason with hierarchical, chunked representations of its world and itself.

The v1.0 PoC will thus serve as a robust foundation for the next stage of AGI research, having demonstrated a critical transition from the principles of pure engineering to the engineering of meaning itself.

Works cited

1. Ultraintelligence Implementation Detailed Breakdown_.pdf
2. What is meta-reasoning in AI? - Milvus, accessed on October 1, 2025, https://milvus.io/ai-quick-reference/what-is-metareasoning-in-ai
3. Improve AI Thinking with Meta-Reasoning Techniques - Relevance AI, accessed on October 1, 2025, https://relevanceai.com/prompt-engineering/improve-ai-thinking-with-meta-reasoning-techniques
4. Enhancing Reflective Practice in the Age of AI - Teaching and Learning | University of Saskatchewan, accessed on October 1, 2025, https://teaching.usask.ca/articles/2025-02-28-reflective-practice-age-ai.php
5. AI Agents that “Self-Reflect” Perform Better in Changing Environments | Stanford HAI, accessed on October 1, 2025, https://hai.stanford.edu/news/ai-agents-self-reflect-perform-better-changing-environments
6. THE IMPACT OF GODEL ON AI - European Scientific Journal, ESJ, accessed on October 1, 2025, https://eujournal.org/index.php/esj/article/view/2930/2759
7. Kurt Godel and The Limits of AI | Ben Garrido's Author Page, accessed on October 1, 2025, https://bengarrido.com/kurt-godel-and-the-limits-of-ai/
8. What happens when your AI goes rogue? - YouTube, accessed on October 1, 2025, https://www.youtube.com/watch?v=CDHgy4j6QLw
9. When AI Goes Rogue: Blackmail, Shutdowns, and the Rise of High-Agency Machines, accessed on October 1, 2025, https://www.youtube.com/watch?v=k9h2-IEf9ZM
10. Knowledge representation and reasoning - Wikipedia, accessed on October 1, 2025, https://en.wikipedia.org/wiki/Knowledge_representation_and_reasoning
11. Knowledge Representation in AI - GeeksforGeeks, accessed on October 1, 2025, https://www.geeksforgeeks.org/artificial-intelligence/knowledge-representation-in-ai/
12. What is Knowledge representation and reasoning | AI Basics - Ai Online Course, accessed on October 1, 2025, https://www.aionlinecourse.com/ai-basics/knowledge-representation-and-reasoning
13. Can AI Make Analogies? - Bioengineer.org, accessed on October 1, 2025, https://bioengineer.org/can-ai-make-analogies/
14. Why Can'T GPT Think Like Us? Study Reveals AI Limitations In Analogical Reasoning, accessed on October 1, 2025, https://quantumzeitgeist.com/why-cant-gpt-think-like-us-study-reveals-ai-limitations-in-analogical-reasoning/
15. Conceptual blending - Wikipedia, accessed on October 1, 2025, https://en.wikipedia.org/wiki/Conceptual_blending

16. Conceptual Blending in EL++ - Digital CSIC, accessed on October 1, 2025, https://digital.csic.es/bitstream/10261/156050/1/CEUR-WS%28DL2016%29V.1577.pdf
17. Creativity and Artificial Intelligence: A Conceptual Blending Approach - ResearchGate, accessed on October 1, 2025, https://www.researchgate.net/publication/332711522_Creativity_and_Artificial_Intelligence_A_Conceptual_Blending_Approach
18. Computational Game Creativity - AIDA - AI Doctoral Academy, accessed on October 1, 2025, https://www.i-aida.org/course/computational-game-creativity/
19. Computational creativity - Wikipedia, accessed on October 1, 2025, https://en.wikipedia.org/wiki/Computational_creativity
20. Artificial consciousness - Wikipedia, accessed on October 1, 2025, https://en.wikipedia.org/wiki/Artificial_consciousness
21. AI Consciousness Detection: 2025 Tools for Identifying Self-Aware Research Systems, accessed on October 1, 2025, https://editverse.com/ai-consciousness-detection-2025-tools-for-identifying-self-aware-research-systems/
22. Can AI Be Self-Aware? - Unaligned Newsletter, accessed on October 1, 2025, https://www.unaligned.io/p/can-ai-be-self-aware